

Appendix 2

Wenzhou-Kean University Graduation Project/Thesis Approval Form

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Project/Thesis topic:

In GenAI, How Perceived Ease of Use, Perceived Interactivity, and Perceived Personalization Affect Purchase Intention. _____

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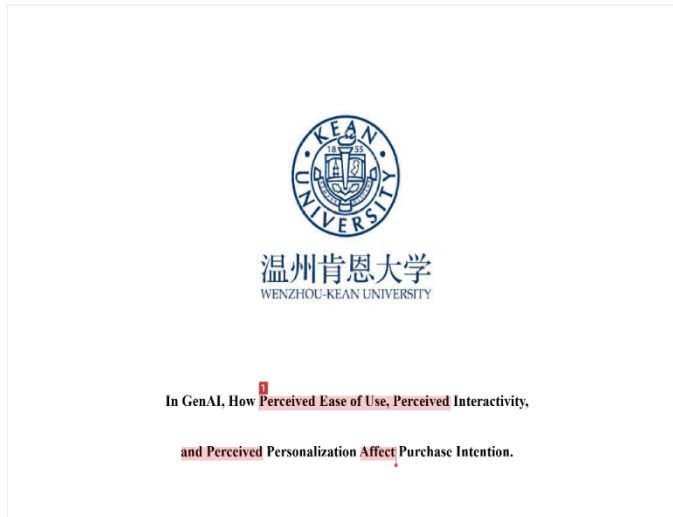
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Appendix 5

Wenzhou-Kean University Graduation Project/Thesis Approval Form

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In GenAI, How Perceived Ease of Use, Perceived Interactivity, and Perceived Personalization Affect Purchase Intention. for the period indicated below **(choose one only)**.

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_____ Chen Siyu _____

Name of the Student (Print)

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Student ID

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Date

_____ Chen Siyu _____

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_____ Ataulah Kiani _____

Name of the Advisor (Print)

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Signature



温州肯恩大学
WENZHOU-KEAN UNIVERSITY

**In GenAI, How Perceived Ease of Use, Perceived Interactivity, and Perceived
Personalization Affect Purchase Intention.**

In Partial Fulfillment of the Requirements For the Bachelor of Science in Management

(Business Analytics Option)

by

Chen Siyu

1235370

December 2, 2025

**In GenAI, How Perceived Ease of Use, Perceived Interactivity,
and Perceived Personalization Affect Purchase Intention.**

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Abstract

Generative Artificial Intelligence (GenAI) has revolutionized the way the digital marketing and customer service industries operate, enabling interactions with customers in a completely new way and providing a tailored shopping experience on an entirely new level. This project focuses on studying how unique, and in this case, the key characteristics of GenAI, such as Perceived Ease of Use (PEOU), Perceived Interactivity (PI), and Perceived Personalization (PP), shape consumers' Purchase Intention (PINT). Using the S-O-R model, this research proposes these characteristics as stimuli that affect BE and CE, and consequently purchase intention, with mediation. Furthermore, Brand Identification (BI) and AI Familiarity (AF) were included as moderators in the model. The research surveyed 320 consumers with prior experience using GenAI chatbots from 4 major regions in China and analyzed the results with SEM and moderation analysis. Confirmed findings for the study indicated that PEOU, PI, and PP had a significant positive impact on CE and BE, which partially or fully mediated their effects on PINT. In particular, the AF negatively moderates the relationship between PEOU and Engagement. Positively, BI moderates the relationship between PP and Engagement. The findings contribute value to the theoretical framework of GenAI and psychology, providing strategies for designing marketing interfaces driven by AI.

Keywords: Generative Artificial Intelligence (GenAI); Perceived Ease of Use; Perceived Interactivity; Perceived Personalization; Behavioral Engagement; Cognitive Engagement; Purchase Intention; Stimulus-Organism-Response (S-O-R) Framework; Brand Identification; AI Familiarity

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1. Introduction

An increasing number of companies are engaging with Generative Artificial Intelligence (GenAI) for marketing and customer relationship management (Kumar & Shankar, 2025). AI-powered GenAI customer service chatbots can offer highly personalized customer recommendations, provide real-time customer support, and offer complex shopping experiences through human-level text and solution generation (Gursoy et al., 2023; Sivathanu & Pillai, 2025). GenAI chatbots are also being utilized by retail giants such as Amazon and Alibaba to customize offerings and facilitate purchases (Bansal et al., 2024; Prentice & Nguyen, 2021a). These GenAI systems are also designed to be an interactive and responsive type of AI stimulus, likely to affect the dynamics of customer interactions and the decision-making process (S. Wang et al., 2025).

As artificial intelligence (AI) is increasingly raised in the workplace. A comprehensive study of user experiences connected with these systems is essential. Within the Technology Acceptance Model (TAM), perceived ease of use refers to the extent to which a user feels that using a particular system would be effortless (Davis, 1989). PEOU would be an obvious immediate determinant of perceived attitude towards the advantages of the IS. A certain efficacy exists with respect to learned literature, which suggests that ease of use affects the perceived usefulness of educational types of literature that ChatGPT relates to (S. Singh & Paiva, 2025). If individuals find GenAI systems effortless and can pose small queries, request minor changes, or articulate complex matters clearly, it is a significant achievement. Cognitive ease when the applications work generates this stimulus, which persuades users to use the system. (Gursoy et al., 2023; Theresiawati et al., 2025).

Perceived interactivity is crucial in shaping user engagement with Gen AI systems. Perceived interactivity is defined as the experience of someone interacting with technology as a form of communication that is two-way, reciprocal, and responsive (F. Wang et al., 2025). In the case of GenAI chatbots, interactivity does not stop at functional input and output. To interact means to experience emotions. GenAI systems that respond empathetically, evaluating users'

emotional states and adjusting their tone or context, can evoke pleasure, enjoyment, and a sense of connectedness (Gao et al., 2023; Prentice & Nguyen, 2021b). This emotional response is important because it shifts the interaction away from task-focused information retrieval, transforming the encounter into something socially rich and engaging. This emotional engagement is motivating. It makes the interaction rewarding and captivates users (Fan, 2025).

Another vital variable for understanding AI-human interaction is perceived personalization. Perceived personalization refers to the extent to which a consumer views the output generated through AI as customized to their preferences, needs, and the context of the situation (Gao et al., 2023). This becomes the leading value proposition of GenAI, as AI systems shift toward personalization and customization, rather than the simpler, rule-based systems of recommendations. Advanced GenAI systems can perform predictive analyses through AI and data generated by the user to create new content, recommendations, and solutions (Hu & Shao, 2025; Kumar & Shankar, 2025). The perception of a service being made for a consumer satisfies the core consumer need for relevance and uniqueness. Stimuli with a sense of personalization become more potent. Personalized communications signal a strong understanding of customers, which increases perceived relevance, exemplified in the utilization of the interaction (S. Wang et al., 2025).

The cognitive and behavioral aspects of customer engagement all hold tremendous value when studying the Gen AI systems. Cognitive engagement, in contrast, pertains to the mental concentration and effort users dedicate to their interactions with the AI system. Customer engagement is a state of mind that is motivational, accompanied by the necessary and appropriate levels of cognitive, emotional, and behavioral activity in relation to a brand and its interactions. This is crucial in interactions with AI-human stimuli, which must be translated to achieve the required outcomes (Elmashhara et al., 2024). Engagement behavior refers to the active signs, actions, energies, and behaviors that customers demonstrate and display when interacting with GenAI. Indicators include counting the duration of conversations, the amount and complexity of questions asked, the extensiveness of the AI conversation, and the interaction with hyperlinks included in the AI recommendations (Agarwal et al., 2025).

As far as consumer behavior is concerned. Purchase intention forms a key aspect of the consumer purchasing process, encompassing the consumer's intent and undertakings to acquire a proposed service or product. This intention directly evolves into a purchase, serving as a key indicator in evaluating one's marketing efforts (Negm, 2025).

Brand identification and AI familiarity as moderating variables of how GenAI systems affect customer engagement and purchase intentions. Brand identification is the consumer's sense of oneness and belonging with a brand, such that the consumer's self-definition becomes contingent on the brand (Davvetas & Diamantopoulos, 2017). This existing relationship becomes important as a moderator. To highly identified consumers, the GenAI interface is more than a tool; it is an extension of a brand they trust and appreciate (Ghorbanzadeh et al., 2025). AI familiarity refers to the understanding and prior experience a consumer has regarding AI technologies (Foroughi et al., 2025). Greater familiarity is expected to diminish feelings of falsity — the uncanny and distrustful feelings that can accompany interactions with a non-human entity that conversationally mimics a human. Skeptical users of GenAI technology may, based on the literature, describe the technology as unsettling or even gimmicky (Agarwal et al., 2025; Feng & Sun, 2025). For users with prior interaction, there is an understanding of the basic functionalities of generative AI systems, as well as their limitations. The artificiality of AI technology is less disturbing, and most users prefer its ease, interactivity, and personalization (Arce-Urriza et al., 2025).

GenAI tools are rapidly transforming the digital marketing landscape and significantly impacting customer support and personalized shopping experiences (Mirzayi & Nobari, 2025). There are still challenges to be resolved regarding the adoption of GenAI for shopping decisions. The “sense of falsity” or “falseness” feeling regarding the AI-generated content is one of the most important challenges. Although GenAI offers speedy and efficient content generation, the responses often lack meaning and value, giving rise to a sense of falseness. (Aljarah et al., 2025). Consequently, the perceived value of the GenAI interaction is most important. It must be effortlessly usable, highly valuable, and interactive, and most of all, GenAI has to generate content that is highly individualized to the user in order to lower the

value skepticism and ultimately lead the user to a purchase (S. Wang et al., 2025).

The very essence of GenAI is an interactive and personalized nature. Prior work has noted and studied interactivity and personalization separately (Anderson et al., 2025). Few studies have integrated emotional experience resulting from interaction and the level of output customization to consumer decision-making. Lastly, research on customer engagement and its impact on purchase intention is lacking, especially for GenAI systems (Liu et al., 2025).

In addition, although there appears to be agreement that responses from consumers are not homogeneous, there has been little focus on the internal emotional states of consumers, including their previous connection to the brand (brand identification) (Davvetas & Diamantopoulos, 2017) and their previous knowledge of the technology (AI familiarity) (Foroughi et al., 2025), which has the potential to alleviate feelings of inauthenticity and influence this complex psychological process to some degree. Nevertheless, scholarly discussions continue on whether customer engagement drivers of Generative AI technology effectiveness are functional or whether they build psychological rapport with the branded firm (Kumar et al., 2025). Studies describe the need to understand better the mechanisms through which Generative AI's customer engagement elements translate into client purchasing decisions and consumer behavior (Mogaji & Jain, 2024).

Consequently, this study seeks to identify the GenAI characteristics that influence purchase intention through the mediation of behavioral and cognitive engagement. So three questions are proposed to guide the research:

RQ1. In what ways do ease of use, interactivity, and personalization foster behavioral and cognitive engagement?

RQ2. In what ways do behavioral and cognitive engagement jointly mediate the effect of GenAI characteristics on purchase intention?

RQ3. How do brand identification and AI familiarity moderate this mechanism?

The implications studied contribute to current research in terms of both theoretical contributions and their management. In terms of theoretical work related to the impact of

GenAI, this study integrates information technology (e.g., technology acceptance) from the disciplines of information systems and user psychology. Here, an effort is made to develop a detailed theoretical framework that clarifies the operational relationships between specific GenAI attributes and different engagement pathways, as well as how these pathways are moderated by various user characteristics. In terms of management, the insights make it possible for brands to optimize GenAI systems according to dimensions of ease of use, interactivity, and high-involvement personalization. This can lead to better and more memorable shopping experience designs, resulting in more trusted user-brand connections on important occasions and improved overall value exchange in sales promotions, ultimately leading to better conversion rates.

The S-O-R (Stimulus-Organism-Response) framework is from environmental psychology (Jacoby, 2002). It is a strong theory that helps in understanding how external environmental aspects affect internal conditions and produce behaviors. Here, different elements of a given environment represent stimuli (S) that impact a person's cognitive and emotional states (O) and ultimately, the response behaviors (R). The S-O-R model has been utilized in both retail and online contexts whereby elements of a given environment (e.g., atmospherics, website layout, and customer service) affects the internal evaluations and emotional states of the consumers, which in turn, causes approach and avoidance behaviors such movement towards purchase, loyalty, and expenditures (Eroglu et al., 2003; Jacoby, 2002). In the new context of interactions between humans and AI, the S-O-R model is increasingly applied to explain how the characteristics of AI technology act as critical stimuli that affect the cognitive and emotional processes of consumers (Kumar & Shankar, 2025; Prentice & Nguyen, 2021a).

Following this framework, this study defines the relationships between variables. The Stimulus Component is the ease of use, interactivity, and personalization (emotional experience) of the conversational GenAI interface. These design features are the first stimuli a consumer engages with. The Organism component encapsulates the psychological phenomena that these stimuli trigger. This is defined operationally through the dual-dimensional customer engagement psych

construct, where behavioral engagement is defined as observable actions, and the cognitive engagement focuses on the mental effort invested in GenAI. Purchase intention is the behavioral Response the consumer outputs in this context. In addition, the model accounts for the moderating effects of brand identification and AI familiarity on the relationships between stimuli and organisms. This acknowledges that not all consumers perceive and process these stimuli in the same way.

Perceived Ease of Use, Interactivity, and Personalization are three variables that influence two mediator variables: Behavioral Engagement and Cognitive Engagement. These two mediating variables influence Purchase Intention. Brand Identification and AI Familiarity function as moderators in this framework. These variables can alter the strength or weakness of, and the nature of, the relationships between the framework's main components.

This study examines the mediating role of Behavioral and Cognitive Engagement. They are on the impact of Perceived Ease of Use, Interactivity, and Personalization on Purchase Intention. The second step of analysis involves moderation analysis, which assesses the extent to which the moderators, Brand Identification and AI Familiarity, affect the proposed relationships in the framework.

The relationships described are captured in the conceptual framework demonstrated in Figure 1. Figure 1 depicts the integrated effect on consumer responses to GenAI.

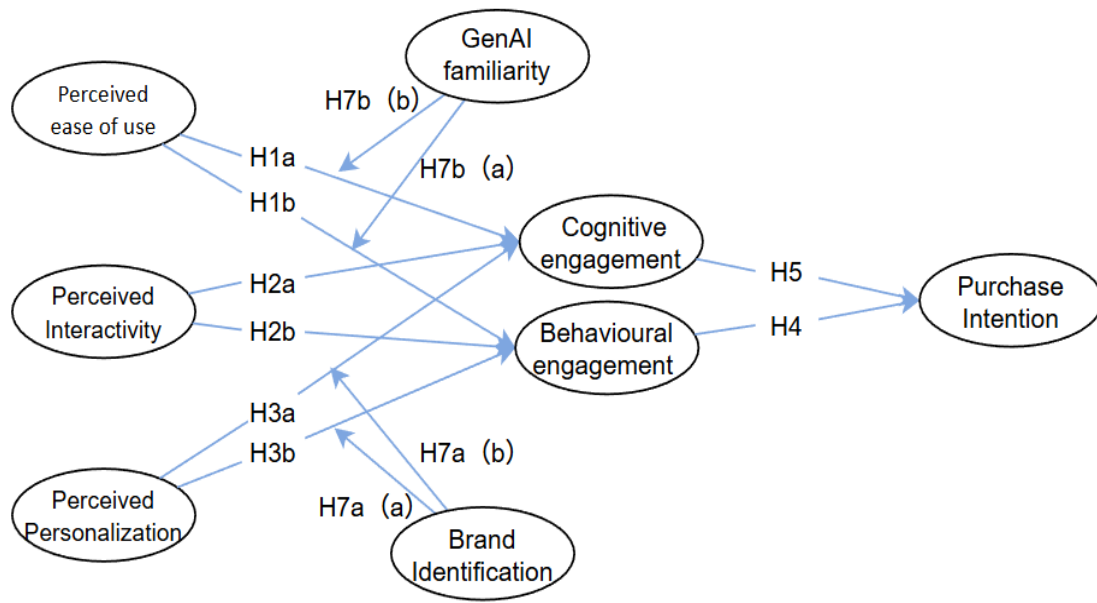


Figure 1. Conceptual Model

2. Literature Review

2.1 Theory underpinning

The Stimulus-Organism-Response (S-O-R) framework is one of the first theories explaining how environmental factors shape the inner workings of a consumer and drive consumer action (Arora et al., 2020). Regarding human-computer interaction, the S-O-R framework posits that a system's technological features (Stimulus) influence a user's thoughts and emotions (Organism), ultimately leading to a corresponding action (Response) (Premathilake et al., 2025). This is especially true for conversational GenAI. Given the digital world's interactivity and psychological engagement of S-O-R stimulated technologies, more specific engagement theories are incorporated. Consumer Engagement Theory (CET) is a multidimensional construct. It provides a complete definition of the "Organism" as a state of engagement where the user deploys cognitive, emotional, and behavioral resources to interact with the brand (Hollebeek et al., 2023).

The S-O-R framework is significant due to its widespread application in various areas of digital marketing, including websites, social media, and e-commerce (Ligaraba et al., 2023). To analyze the effects of digital touchpoints on consumer decisions, the S-O-R framework has been used most commonly in highly interactive and personalized environments (Erensoy et al., 2024). However, with more complex and relational AI-driven interfaces, companies should learn how to make deep and meaningful interactions (Vafaei-Zadeh et al., 2025).

As an extension of the S-O-R paradigm, Consumer Engagement Theory (CET) describes psychological processing through two primary pathways: behavioral and cognitive pathways (Hoang et al., 2023). CET describes the focus of a consumer's mental attention (cognitive engagement) and the extent of their actions (behavioral engagement) during an interaction, both of which are important in the attainment of enduring behavioral outcomes such as loyalty and purchase intention (Vongkulluksn et al., 2022). CET has applications for firms using Generative AI because it can help a firm understand how technology's attributes become psychological ties and commercial outcomes (Marchena Sekli & Portuguese-Castro, 2025).

Using CET, firms can attract more attention. They can develop systems for extracting cognitive and behavioral resources during the purchase process (Hussin & Wahid, 2023).

2.2-1 Purchase Intention

Purchase intention (PI) refers to a consumer's conscious plan and willingness to make a particular purchase (Baltaci et al., 2024). Understanding what drives PI is crucial for brands and buyers in the rapidly evolving digital marketplace, and new technologies like conversational Generative AI are being developed. Purchase intention is a strong predictor of actual buying behavior (Prasanna & Kushwaha, 2025). As GenAI chatbots and assistants become increasingly important consumer brand interfaces, there is a surge in interest in marketing research on the psychology of consumer purchasing intent and how these technologies influence it (Pathak et al., 2025). Hence, this research advances the position of purchase intention as the foremost variable in assessing the effectiveness of consumer interactions using GenAI.

2.2-2 Perceived Ease of Use and Consumer Engagement

Perceived Ease of Use (PEOU) refers to how a user perceives the effort involved in using a system (Davis, 1989; Topsakal, 2025). PEOU is one of the key constructs in the Technology Acceptance Model (TAM) and a determinant of technology adoption. Conversational GenAI tools provide cognitive assistance to users, and GenAI-powered tools that are more user-friendly and responsive are more likely to alleviate cognitive overload (Qian, 2025). When users do not struggle with the interface, they can concentrate on the interaction, which increases BE levels by extending session duration and exploration. In contrast, CE levels increase as attention is freed from the tools used to access the material (S. Singh & Paiva, 2025). Thus, the study hypothesizes that PEOU is a significant predictor of the engagement state.

H1a. Perceived Ease of Use has a significant and positive impact on Behavioral Engagement.

H1b. Perceived Ease of Use has a significant and positive impact on Cognitive Engagement.

2.2-3 Perceived Interactivity and Consumer Engagement

Perceived Interactivity is the extent to which users perceive communication with a given technology as reciprocal, responsive, and synchronized (Joo & Yang, 2023). Regarding conversational GenAI, the expectation is that the agent fully participates in a conversation, appreciates contextual nuances, responds instantly, and provides relevant responses to prompts (Joo & Yang, 2023). High interactivity serves as a direct driver of consumer engagement. By stimulating curiosity, the interactivity of a system to the point of CE will facilitate deeper information processing and sustained mental focus on the conversation involving the brand. Conversely, these systems will actively engage users in behavioral engagement (BE) by motivating them to ask questions, extending the duration of their interaction, and exploring various product options (Agarwal et al., 2025; Gao et al., 2023). Thus, the study hypothesizes:

H2a. Perceived Interactivity has a significant and positive impact on Behavioral Engagement.

H2b. Perceived Interactivity has a significant and positive impact on Cognitive Engagement.

2.2-4 Perceived Personalization and Consumer Engagement

The extent to which a customer feels that the content and results produced by an AI system are tailored to their specific tastes, requirements, and previous actions is termed Perceived Personalization (Hu & Shao, 2025). Conversational GenAI utilizes available user information and ongoing interactions to deliver highly personalized recommendations and tailored advice regarding a customer's purchasing decision (Phuong et al., 2025). This experience of having something created for an individual is a profound motivator of engagement. The more customers sense that a brand appreciates their individuality, the more they are willing to invest in that brand (Gao et al., 2023). The relevance of the interaction and the value of the experience are deepened psychologically:

H3a. Perceived Personalization has a significant and positive impact on Behavioral Engagement.

H3b. Perceived Personalization has a significant and positive impact on Cognitive Engagement.

2.2-5 Consumer Engagement and Purchase Intention

Consumer Engagement, particularly in terms of its behavioral and cognitive aspects, has a substantial impact on business performance. Behavioral Engagement (BE) takes the form of repeated visits, content sharing, and prolonged periods of interaction, which enhance familiarity and deepen emotional and psychological investment in a brand, ultimately leading to a purchase (Hospel et al., 2016; Onofrei et al., 2022). Cognitive Engagement (CE), which involves a consumer's focused mental attention and brand-related thinking, drives a deeper understanding, stronger brand associations, and more positive brand evaluations, thereby fostering purchase intention (Li & Lajoie, 2022).

Researchers found CE and BE to be statistically significant in influencing purchase intention. It is therefore that purchases are a natural culmination of the relationship when a consumer's brand relational bond strengthens (Agarwal et al., 2025; Greene, 2015).

H4. Behavioral Engagement has a significant and positive impact on Purchase Intention.

H5. Cognitive Engagement has a significant and positive impact on Purchase Intention.

2.2-6 The Mediating Role of Consumer Engagement

The proposed mediating role of engagement is theoretically grounded in the Stimulus-Organism-Response (S-O-R) model and the Cognitive-Affective-Behavioral framework. The three attributes of GenAI are explicitly defined in this study as the main external stimuli: Perceived Ease of Use, Perceived Interactivity, and Perceived Personalization. Behavioral Engagement (BE) and Cognitive Engagement (CE) are the core organismic states. Past literature supports this, as it identifies engagement as the essential psychological mechanism through which marketing stimuli are translated into outcomes (Harmeling et al., 2017). Finally, Purchase Intention is positioned as the ultimate behavioral response in this causal chain.

Purchase Intention is used as a final response in this causal structure. The mediation statements indicate that the effect of technological stimuli in the final result will not be merely direct but is mostly transmitted through these internal engagement states. In this regard, for instance,

research into interactive media shows that perceived interactivity enhances purchase intention, as it improves both cognitive and behavioral engagement among users (Joo & Yang, 2023). Equally, research into personalized systems shows that the impact of personalization on behavioral results is fully mediated by enhanced levels of user engagement (Teepapal, 2025). Consequently, mediation is proposed in the following hypotheses:

H6a. Behavioral Engagement and Cognitive Engagement mediate the relationship between Perceived Ease of Use and Purchase Intention.

H6b. Behavioral Engagement and Cognitive Engagement mediate the relationship between Perceived Interactivity and Purchase Intention.

H6c. Behavioral Engagement and Cognitive engagement mediates the relationship between Perceived Personalization and Purchase Intention.

2.2-7 The Moderating Role of Brand Identification and AI Familiarity

The potential strength of each proposed relationship may vary according to individual consumer characteristics. Brand Identification, the degree to which a consumer possesses a psychological sense of belonging and connection for a brand (Bhattacharya & Sen, 2004), is especially critical to amplifying the engagement-purchase pathway. Such a consumer is more likely to have a positive engagement experience with a brand's GenAI agent and to have greater motivation to purchase.

On the other hand, AI Familiarity, which describes a consumer's prior knowledge and comfort with AI, especially functions in the initial stages of interaction. Consumers with low familiarity may hold what (Aljarah et al., 2025) describes as a "sense of falsity." This skepticism could diminish the impact of GenAI features on engagement and neutralize the effect of a well-crafted agent, designed to ease interaction, promote responsiveness, and personalize the experience. For more AI Familiar consumers, appreciation or responsiveness to these designed attributes is more likely.

H7a. Brand Identification positively moderates the relationship between Perceived

Personalization and Engagement ((a) Behavioral Engagement and (b) Cognitive Engagement), such that the relationships are stronger for consumers with high brand identification.

H7b. AI Familiarity positively moderates the relationship between Perceived Ease of Use and Engagement, and Engagement ((a) Behavioral Engagement and (b) Cognitive Engagement), such that the relationships are stronger for consumers with high AI familiarity.

3. Research Methodology

3.1 Data analysis approach

This quantitative study utilizes a cross-sectional design. The data were analyzed using Version 26 of the Statistical Package for the Social Sciences (SPSS) and the Analysis of Moment Structures (AMOS).

Raw data were screened and cleaned to identify missing values and outliers. After assessing the quality of the dataset, an Exploratory Factor Analysis (EFA) was performed in SPSS to evaluate the factors that comprise the scales and assess the overall appropriateness of the measurement scales.

This was followed by Confirmatory Factor Analysis (CFA) in AMOS to evaluate the measurement model that was established a priori. Model fit was assessed based on a set of absolute (e.g., χ^2/df , RMSEA), incremental (e.g., CFI, TLI), and parsimony (PNFI) goodness of fit indices. The measurement model robustness was qualified by testing the reliability of validity (internal consistency reliability, CR, and Cronbach's Alpha, convergent validity AVE & SFL, discriminant validity. Furthermore, the indicator reliability of each measured item was checked through Squared Multiple Correlations (SMC).

After confirming the measurement model, I tested the hypothesized model using AMOS software with Structural Equation Modeling (SEM). Being an advanced multivariate technique, SEM allows the simultaneous examination of complex relationships between latent constructs posited in the proposed research model. Finally, to test the proposed mediation effects, I followed a bootstrapping confidence interval procedure proven superior to the traditional Baron and Kenny method as well as the Sobel test in testing the mediation effect

3.2 Sampling

The data for this study were collected in three major economic provinces and one municipality in China, including Zhejiang, Anhui, Jiangsu, Shanghai, and Guangdong. These regions were chosen to reflect different economically dynamic consumer markets, making them relevant to

the findings in an early digital commerce market in China. Ensuring the relevance of the findings to key digital commerce hubs in China.

For primary data collection, a multi-channel online survey was utilized. The survey questionnaire was distributed through online panel services and social media platforms targeting users in the designated areas, a strategy commonly employed in current consumer behavior studies (Lehdonvirta et al., 2021). To qualify for inclusion, respondents were required to have prior experience with the use of conversational Generative AI (e.g., chatbots, AI assistants) in relation to product inquiries and online shopping.

The data collection process extended over a period of six weeks. A total of 321 questionnaires were gathered at the beginning. After an intensive data screening process to eliminate responses that contained straight-lining, inconsistent answers, or missing values, 320 usable responses remained for final data analysis, resulting in an effective response rate of 99.7%.

3.3 Data Collection

The final sample of 320 respondents evidences a high degree of relevance to the research environment. All valid respondents (100%) reported that they had used AI, and virtually all (99.7%) indicated that they had had experience with generative AI chatbots, such as GPT, DeepSeek, Doubao, or ERNIE Bot. In addition, there was a strong majority (82.2%) that had either been exposed to or used a Gen AI chatbot while shopping online, indicating the appropriateness of the sample for investigating AI-oriented consumer behaviors.

The demographic characteristics of the sample showed a reasonably balanced representation in terms of gender, with 50.9% males ($n = 163$) and 49.1% females ($n = 157$). In terms of generational distribution, the largest group was born between 1997 and 2010 (45.3%, $n=145$), followed by those born between 1980 and 1996 (29.1%, $n=93$), which can be regarded as the core digital commerce demographic. The younger respondents born between 1939 and 1979 accounted for 14.7% ($n=47$), while those born in 2010 or later accounted for 10.9% ($n=35$).

In terms of socioeconomic characteristics, the sample revealed a relatively high-income consumer base, with a significant proportion (62.5%, n = 200) reporting a monthly disposable income of over ¥5,000. The group represented by the income bracket of ¥3,000-5,000 accounted for 24.1% (n=77), while the lower monthly income bracket (below ¥3,000) represented 153 13.4% (n=43) of the sample population. The details on sample demographics are provided in Table 1.

Classification	Description	Frequency	Percentage(%)
Gender	Male	163	50.9
	Female	157	49.1
	Total	320	100.0
Age	46 and above	47	14.7
	29 – 45	93	29.1
	15 – 28	145	45.3
	14 or below	35	10.9
	Total	320	100.0
Monthly Disposable Income	Below ¥1500	18	5.6
	¥1500 - 3000	25	7.8
	¥3000 - 5000	77	24.1
	Above ¥5000	200	62.5
	Total	320	100.0
Generative AI Chatbot Experience	Has Experience	320	100.0
	No Experience	0	0
	Total	320	100.0
Gen AI Shopping Encounter	Has Used	263	82.2
	Never Used	57	17.8
	Total	320	100.0

Table 1
Characteristics
of respondents

3.4 Measures

To achieve construct validity for the research model, relevant literature was used to establish the reflective scales for each construct. The questionnaire consisted of the seven-point Likert scale (from 1 Strongly Disagree to 5 (Strongly Agree)).

Items in the questionnaire for the independent variables were taken from: Davis (1989); Singh & Paiva (2025) for Perceived Ease of Use (4 items), Gao et al. (2023) for Perceived Interactivity (6 items), and Gao et al. (2023) for Perceived Personalization (5 items).

For the mediated variables: Agarwal et al. (2025); Fang et al. (2017); McLean & Wilson (2019) for Behavioral Engagement (3 items) and Agarwal et al. (2025); McLean & Wilson (2019); O'Brien et al. (2018) for Cognitive Engagement (5 items).

There were also scales referenced for the moderating variable: Brand Identification used 5 items from Davvetas & Diamantopoulos (2017); Stokburger-Sauer et al. (2012) measured AI Familiarity with four items based on the scale of Feng & Sun (2025).

For the dependent variable Purchase Intention, which was measured using 3 items from Arachchi & Samarasinghe (2024), Prentice et al. (2019) were utilized.

4. Data Analysis and Results

4.1 Measurement and Pre-testing

This research utilizes established measurement scales to evaluate Perceived Ease of Use (PEOU), Perceived Interactivity (PI), Perceived Personalization (PP), Behavioral Engagement (BE), Cognitive Engagement (CE), AI Familiarity (AF), Brand Identification (BI), and Purchase Intention (PINT). These are those who were familiar with the past information systems and marketing research. The participants' answers were gathered using a 5-point Likert scale.

Perceived Ease of Use (PEOU) was measured using a four-item scale adapted from Davis (1989); Singh & Paiva (2025); Cronbach's $\alpha = 0.815$. Perceived Interactivity (PI) was measured using a six-item scale from Gao et al. (2023); Cronbach's $\alpha = 0.884$. Perceived Personalization (PP) was measured using a five-item scale from Gao et al. (2023); Cronbach's $\alpha = 0.867$. A three-item scale assessed Behavioral Engagement (BE) from Agarwal et al. (2025); Fang et al. (2017); McLean & Wilson (2019); McLean & Wilson (2019); $\alpha = 0.838$. Cognitive Engagement (CE) was measured using a five-item scale from Agarwal et al. (2025); McLean & Wilson (2019); O'Brien et al. (2018); $\alpha = 0.866$. AI Familiarity (AF) was measured using a four-item scale from Feng & Sun (2025); $\alpha = 0.857$. Brand Identification (BI) was measured using a five-item scale from Davvetas & Diamantopoulos (2017); Stokburger-Sauer et al. (2012); $\alpha = 0.870$. Purchase Intention (PINT) was measured using a three-item scale from Arachchi & Samarasinghe (2024); Prentice et al. (2019); $\alpha = 0.821$.

All the measurement scales and factor loadings are in Table 2.

Latent variable	Factor loading	Cronbach's alpha (α)
PEOU		0.815
PEOU1: Using generative AI is innovative compared to other tools I use for shopping	0.806	
PEOU2: Using generative AI offers new and creative ways to support my shopping decisions	0.811	
PEOU3: Using generative AI in my shopping process is a groundbreaking idea	0.816	
PEOU4: Using generative AI enhances my shopping experience	0.811	
PI		0.884
PI1: Generative AI chatbot were very slow in responding to my request	0.888	
PI2: The generative AI chatbot can accurately provide me with the product	0.817	
PI3: When I encounter a shopping-related problem, the generative AI chatbot can provide me with a solution	0.699	
PI4: The generative AI chatbot can effectively collect customer feedback	0.832	
PI5: The generative AI chatbot makes me feel like I'm having a real conversation	0.776	
PI6: The generative AI chatbot does not encourage users to talk back	0.999	
PP		0.867
PP1: The generative AI chatbot stores my preferences and offers me extra services based on my preferences	0.860	
PP2: The generative AI chatbot does a pretty good job guessing what kinds of products I might want and making suggestions	0.837	
PP3: The generative AI chatbot know what I want	0.930	
PP4: The generative AI chatbot setup can be personalized to my needs	0.951	
PP5: The service provided by the generative AI chatbot is customized exactly to my question	0.953	
BE		0.838
BE1: I often actively engage with the generative AI chatbot during my shopping	0.860	
BE2: I frequently explore new products, activities, or promotions by using the generative AI chatbot	0.915	
BE3: I explore and interact with various features and recommendations provided by the generative AI chatbot.	0.733	
CE		0.866
CE1: I am highly focused on my interactions with the generative AI chatbot	0.790	
CE2: I often find myself engaged in problem-solving or decision-making processes with the help of the generative AI chatbot	0.835	
CE3: I generally do not need to invest a lot of mental effort in processing and understanding the information provided by the generative AI chatbot	0.986	

CE4: I feel that my interactions within the generative AI chatbot contribute to my shopping decisions or purchasing knowledge	0.986	
CE5: I find myself deeply engaged for a long duration when using the generative AI chatbot in a single session	0.863	
AF		0.857
AF1: I have had experience interacting with AI	0.961	
AF2: I am familiar with AI-generated content (texts, audio, visuals, etc.)	0.900	
AF3: I am familiar with AI-powered applications or tools in my daily life (e.g., voice assistants, recommendation systems).	0.900	
AF4: I feel comfortable discussing the capabilities and limitations of AI.	0.956	
BI		0.870
BI1: Through interactions with the Gen AI chatbot, I feel a strong sense of belonging to this brand.	0.920	
BI2: The Gen AI chatbot helps me identify strongly with this brand.	0.949	
BI3: Through the Gen AI chatbot, I can believe that this brand embodies.	0.887	
BI4: Due to the Gen AI chatbot, this brand is like a part of me.	0.751	
BI5: Through the Gen AI chatbot, this brand has a great deal of personal meaning to me.	0.749	
PINT		0.821
PINT1: I would purchase products or services through Gen AI chat	0.961	
PINT2: The probability that I would consider buying through Gen AI chat is high	0.859	
PINT3: My willingness to purchase through Gen AI chat is strong	0.833	

Table 2

Exploratory factor analysis

4.2 Data Analysis

Initially, an Exploratory Factor Analysis (EFA) was performed to validate the measurement quality of the scales. In accordance with (B. Singh & Verma, 2024), all items were retained since they met the required cut-off value for item inclusion, with all factor loadings surpassing 0.699 (see Table 2). Additionally, the results of the t-test revealed no significant differences, indicating that non-response bias was not a concern.

Another approach that can be used to deal with Common Method Bias (CMB) concerns is the correlation matrix method (Pavlou et al., 2006). From Table 3, it is evident that all construct inter-correlation values were below 0.80, with the highest correlation observed at 0.484

between Behavioral Engagement and Cognitive Engagement. That means there is no significant CMB in this data.

	PEOU	PI	PP	BE	CE	AF	BI	PINT
PEOU	1							
PI	.365**	1						
PP	.315**	.215**	1					
BE	.477**	.392**	.355**	1				
CE	.396**	.359**	.310**	.484**	1			
AF	.369**	.230**	.322**	.367**	.353**	1		
BI	.424**	.271**	.366**	.449**	.373**	.365**	1	
PINT	.380**	.251**	.258**	.429**	.299**	.354**	.407**	1
Summary statistics								
Number of items	4	6	5	3	5	4	5	3
Mean	3.7578	3.7844	3.6250	3.7719	3.4704	3.6578	3.6913	3.6427
SD	0.90255	0.93708	0.94627	1.01042	0.64555	0.98981	0.91668	0.96604
Note(s): N = 320; SD = Standard Deviation;								
PEOU = Perceived Ease of Use; PI = Perceived Interactivity; PP = Perceived Personalization; BE = Behavioral Engagement; CE = Cognitive Engagement; AF = AI Familiarity; BI = Brand Identification; PINT = Purchase Intention;								
*: $p < 0.05$, **: $p < 0.01$, ***: $p < 0.001$;								
** Correlation is significant at the 0.01 level (2-tailed);								
Correlation is significant at the 0.05 level (2-tailed)								
Source: Authors' estimation								

Table3.

Descriptive statistics
And construct
intercorrelations

4.3 Measurement Model, Reliability, and Validity Analysis

In addition, the results of the reliability and validity tests must be valid; for this to be possible, the standardized factor loadings of the items on their corresponding constructs were all higher than 0.699 (Table 2). The Cronbach's Alpha (α) values for all constructs were greater than 0.821. Overall, the reliability and validity of our measurement model can be considered very high. As shown in the bottom value of Table 4, all the constructions have Composite Reliability (CR) values ranging from 0.877 to 0.962. Additionally, every Average Variance Extracted (AVE) measured is more than 0.658. Collectively, these findings provide strong evidence of

convergent validity (Hauser & Goldberger, 1971; Møller et al., 2005). Discriminate validity is also present in the findings. As shown in Table 4, the square root of AVE for each construct surpassed its correlation with other constructs (off-diagonal values in Table 3). Moreover, each construct's AVE was greater than its Maximum Shared Variance (MSV). These results provide strong evidence for the discriminant validity of this measurement model.

CR	AVE	MSV	ASV	PEOU	PI	PP	BE	CE	AF	BI	PINT
0.885	0.658	0.228	0.269	0.811							
0.934	0.706	0.154	0.163	0.365	0.840						
0.959	0.824	0.134	0.168	0.315	0.215	0.907					
0.877	0.705	0.234	0.315	0.477	0.392	0.355	0.839				
0.953	0.802	0.234	0.242	0.396	0.359	0.310	0.484	0.896			
0.962	0.864	0.136	0.203	0.369	0.230	0.322	0.367	0.353	0.930		
0.931	0.732	0.202	0.257	0.424	0.271	0.366	0.449	0.373	0.365	0.855	
0.916	0.785	0.184	0.210	0.380	0.251	0.258	0.429	0.299	0.354	0.407	0.886

Note(s):CR = composite reliability, AVE = average variance extracted, MSV = maximum shared variance, ASV = average shared variance, PEOU = Perceived Ease of Use; PI = Perceived Interactivity; PP = Perceived Personalization; BE = Behavioral Engagement; CE = Cognitive Engagement; AF = AI Familiarity; BI = Brand Identification; PINT = Purchase Intention;

Source: Authors' estimation

Table4.

Discriminate validity

4.4 Structural Model and Hypotheses Testing

After the instrument model, the Confirmatory structural model was grounded to the proposed relationships, including direct, mediating, and moderating effects. The model fit was assessed with the maximum-likelihood estimation technique. The results showed that the Structural model provided a good fit of the data, all fit indices meet the relevant standard recommended (e.g., χ^2/df , GFI, CFI, RMSEA, etc., as detailed in the subsequent tables), so that the path coefficient is a robust interpretation.

The analysis of the structural model contains two main sections. Firstly, the Direct effects of the independent variables (Perceived Ease of Use, Perceived Interactivity, Perceived

Personalization) on the dependent variable (Purchase Intention), and hypothesized relations with mediators (Behavioral Engagement, Cognitive Engagement) were considered. In the next step, Sobel's test was used to examine the mediating role of Behavioral Engagement and Cognitive Engagement.

Additionally, the effects of our moderating variables, AI Familiarity and Brand Identification, are discussed. A series of mediation models was developed to examine whether the strength of the statistical relationship, from independent variables to mediators of engagement, differs significantly at various levels of moderation.

In Table 5, the detailed outcomes of path coefficients, their significance levels, and the Sobel test for intermediate effects are provided. The results of the moderating effect test are given in Table 6. These results support the research hypotheses, which will be discussed in the following section.

4.5 Testing of hypothesized relationships: (The Roles of Behavioral and Cognitive Engagement as Mediators, and the Moderating Effects of AI Familiarity and Brand Identification)

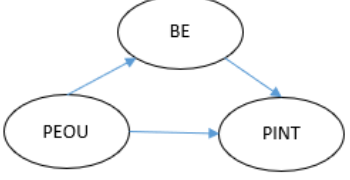
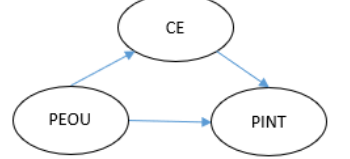
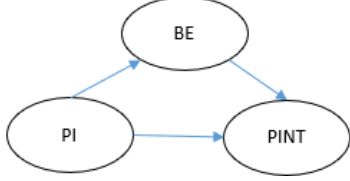
To appropriately examine the mediating and moderating mechanisms, we followed a stepwise statistical approach, which involves integrating path analysis in an SEM framework along with the Sobel test to verify our findings (Mathur et al., 2016; Ro, 2012).

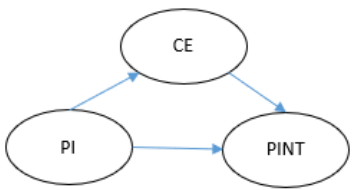
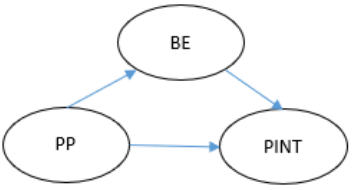
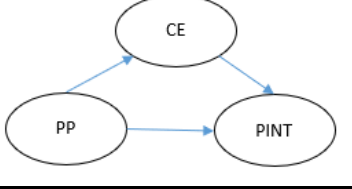
4.5.1 Mediating Effects: Behavioral and Cognitive Engagement

Results from the SEM revealed that all prerequisites for mediation were satisfied. This finding demonstrates the mediating role of BE and CE in the model. More importantly, the results also imply that BE partially mediates the influence of PEOU on PINT ($\beta_{PEOU \rightarrow PINT} = 0.253, p = 0.002$), PI and PINT ($\beta_{PI \rightarrow PINT} = 0.140, p = 0.047$), and CE and PINT ($\beta_{PP \rightarrow PINT} = 0.199, p = 0.004$). However, BE fully mediates the relationship between PP and PINT, as the direct effect becomes non-significant ($\beta_{PP \rightarrow$

PINT = 0.113, $p = 0.096$) once it is introduced. Similarly, CE mediates the relationships between PEOU and PINT, PI and PINT, and PP and PINT.

These indirect effects were also statistically significant, as evidenced by the Sobel test (e.g., $z = 4.403$ for $PP \rightarrow BE \rightarrow PINT$; $z = 3.321$ for $PP \rightarrow CE \rightarrow PINT$), all z -values > the critical value of z -values and p -values < 0.05. These results offer consistent and substantial evidence for the essential mediating roles of BE and CE in translating the effects of AI attributes to purchase intention. The model fit indices for all mediation models (e.g., Model_1: $\chi^2/df = 0.720$, CFI = 1.000; Model_5: $\chi^2/df = 1.221$, CFI = 0.994) were outstanding, indicating an excellent fit to the data (Ro, 2012).

	Pictorial view	SEM results			Sobel results			Remarks
		Path	beta(β)	p-value	z-value	SE	p	
Direct – Model_1		PEOU - PINT	0.380	0.000	-	-	-	Direct association
Mmodel_1 (PEOU - BE - PINT)		PEOU - BE	0.571	0.000	3.954	0.064	0.000	Partial mediation
		BE - PINT	0.369	0.000				
		PEOU - PINT	0.253	0.002				
Mmodel_2 (PEOU - CE - PINT)		PEOU - CE	0.479	0.000	2.324	0.044	0.020	Partial mediation
		CE - PINT	0.180	0.013				
		PEOU - PINT	0.375	0.000				
Direct – Model_2		PI - PINT	0.268	0.000	-	-	-	Direct association
Mmodel_3 (PI - BE - PINT)		PI - BE	0.484	0.000	4.556	0.053	0.000	Partial mediation
		BE - PINT	0.446	0.000				
		PI - PINT	0.140	0.047				

Mmodel_4 (PI - CE - PINT)		PI - CE	0.459	0.000	3.085	0.042	0.002	Partial mediation
		CE - PINT	0.251	0.000				
		PI - PINT	0.239	0.000				
Direct – Model_3		PP - PINT	0.258	0.000	-	-	-	Direct association
Mmodel_5 (PP - BE - PINT)		PP - BE	0.412	0.000	4.403	0.052	0.000	Full mediation
		BE - PINT	0.467	0.000				
		PP - PINT	0.113	0.096				
Mmodel_6 (PP - CE - PINT)		PP - CE	0.365	0.000	3.321	0.038	0.001	Partial mediation
		CE - PINT	0.287	0.000				
		PP - PINT	0.199	0.004				
Note(s): PEOU = Perceived Ease of Use; PI = Perceived Interactivity; PP = Perceived Personalization; BE = Behavioral Engagement; CE = Cognitive Engagement; AF = AI Familiarity; BI = Brand Identification; PINT = Purchase Intention; β = Standardized path coefficient, ρ = Significance level, SE = Standard error Source: Authors' estimation								
Table 5. The mediation results (SEM and Sobel)								

Model	Absolute fit indices				Incremental fit indices		Parsimony fit indices	
	χ^2/df	GFI	RMR	RMSEA	CFI	NFI	AGFI	PCFI
Mmodel_1(PEOU - BE -PINT)	0.720	0.986	0.033	0.000	1.000	0.983	0.977	0.574
Mmodel_2(PEOU - CE -PINT)	1.157	0.971	0.039	0.022	0.995	0.964	0.955	0.635
Mmodel_3(PI - BE -PINT)	1.285	0.966	0.044	0.030	0.992	0.964	0.949	0.632
Mmodel_4(PI - CE -PINT)	0.670	0.951	0.044	0.043	0.979	0.945	0.930	0.670
Mmodel_5(PP - BE -PINT)	1.221	0.972	0.044	0.026	0.994	0.968	0.955	0.604
Mmodel_6(PP - CE -PINT)	1.174	0.968	0.042	0.023	0.994	0.961	0.952	0.659

Note(s): PEOU = Perceived Ease of Use; PI = Perceived Interactivity; PP = Perceived Personalization; BE = Behavioral Engagement; CE = Cognitive Engagement; AF = AI Familiarity; BI = Brand Identification; PINT = Purchase Intention;

X2/df = chi-square/degree of freedom, GFI = goodness of fit index, RMR = root mean residuals, RMSEA = root mean square error of estimation, CFI = confirmatory fit index, NFI = normed fit index, AGFI = adjusted goodness of fit index, PCFI = parsimonious confirmatory fit index

Source(s): Authors' estimation

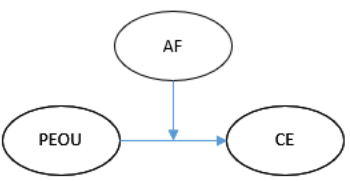
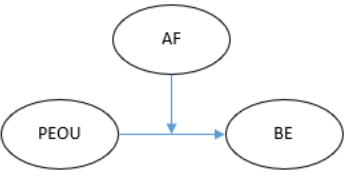
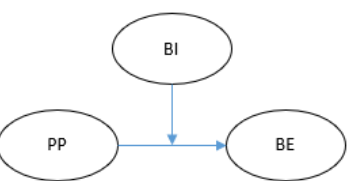
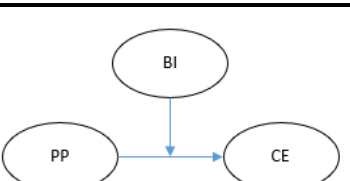
Table 6.
Overall model fit indices

4.5.2 Moderating Effects of AI Familiarity and Brand Identification

The results strongly affirm the proposed moderating effects. Regarding the moderating role of AF, the interaction term PEOU*AF had a significant negative effect on both BE ($\beta = -0.332$, $p < 0.001$) and CE ($\beta = -0.434$, $p < 0.001$). This suggests that the positive effect of PEOU on engagement is significantly weaker among consumers with high AI familiarity. These moderated mediation paths were tested using the Sobel test, and both were found to be significant ($z = -2.804$, $p = 0.005$ for BE; $z = -2.247$, $p = 0.025$ for CE).

For the moderating effect of BI, the product variable PP*BI had a significant positive influence on both BE ($\beta = 0.420$, $p < 0.001$) and CE ($\beta = 0.421$, $p < 0.001$). This demonstrates that the positive effect of PP on engagement is significantly stronger for individuals who are highly identified with the brand (i.e., a high level of Brand Identification). The Sobel test results further supported the significance of these moderated mediation effects ($z = -4.969$, $p < 0.001$ for BE; $z = -4.125$, $p < 0.001$ for CE). All moderated models exhibited good fit with the data (e.g., Model_7: $\chi^2/df = 1.062$, CFI = 0.998; Model_9: $\chi^2/df = 1.210$, CFI = 0.992).

	Pictorial view	SEM results			Sobel results			Remarks
		Path	beta(β)	p-value	t-value	SE	p	
Direct – Model_4		PEOU - CE	0.396	0.000	-	-	-	Direct association
Model_7		PEOU - AF	0.332	0.000	-2.804	0.053	0.005	

(PEOU - AF - BE)		AF - BE	0.200	0.002				Partial moderation
		PEOU - BE	0.393	0.000				
Direct – Model_5		PEOU - BE	0.477	0.000	-	-	-	Direct association
Model_8 (PEOU - AF - CE)		PEOU - AF	0.434	0.000	-2.247	0.059	0.025	Partial moderation
		AF - CE	0.223	0.000				
		PEOU - CE	0.308	0.000				
Direct – Model_6		PP - BE	0.355	0.000	-	-	-	Direct association
Model_9 (PP - BI - BE)		PP - BI	0.420	0.000	-4.969	0.052	0.000	Partial mediation
		BI - BE	0.341	0.000				
		PP - BE	0.164	0.011				
Direct – Model_7		PP - CE	0.310	0.000	-	-	-	Direct association
Model_10 (PP - BI - CE)		PP - BI	0.421	0.000	-4.125	0.8518	0.000	Partial mediation
		BI - CE	0.268	0.000				
		PP - CE	0.161	0.017				

Note(s): PEOU = Perceived Ease of Use; PI = Perceived Interactivity; PP = Perceived Personalization; BE = Behavioral Engagement; CE = Cognitive Engagement; AF = AI Familiarity; BI = Brand Identification; PINT = Purchase Intention;
 β = Standardized path coefficient, p = Significance level, SE = Standard error

Source: Authors' estimation

Table 7.
The moderator results (SEM and Sobel)

Model	Absolute fit indices				Incremental fit indices		Parsimony fit indices	
	χ^2/df	GFI	RMR	RMSEA	CFI	NFI	AGFI	PCFI
Model_7(PEOU - AF - BE)	1.062	0.974	0.047	0.014	0.998	0.968	0.959	0.741
Model_8(PEOU - AF - CE)	1.222	0.962	0.042	0.026	0.991	0.955	0.945	0.660
Model_9(PP - BI - BE)	1.210	0.962	0.040	0.026	0.992	0.958	0.945	0.660
Model_10(PP - BI - CE)	1.046	0.963	0.038	0.012	0.988	0.957	0.949	0.701

Note(s): PEOU = Perceived Ease of Use; PI = Perceived Interactivity; PP = Perceived Personalization; BE = Behavioral Engagement; CE = Cognitive Engagement; AF = AI Familiarity; BI = Brand Identification; PINT = Purchase Intention; χ^2/df = chi-square/degree of freedom, GFI = goodness of fit index, RMR = root mean residuals, RMSEA = root mean square error of estimation, CFI = confirmatory fit index, NFI = normed fit index, AGFI = adjusted goodness of fit index, PCFI = parsimonious confirmatory fit index

Source(s): Authors' estimation

Table 8.
Overall model fit indices

In summary, the combined SEM and Sobel test results provide consistent and robust evidence for the proposed research model, confirming the critical mediating roles of BE and CE, as well as the significant boundary conditions set by AF and BI.

4.6 Moderation analysis

The PROCESS version 4.1 (Hayes, 2017), model 7 was used to identify the moderating effect of AF on PEOU and CE; and model 8 was used to identify the moderating effect of AF on PEOU BE. Model 9 was used to identify the moderating effect of BI on PP and BE; model 10 was used to identify the moderating effect of BI on PP and CE.

This study first examined the moderating effect of Attitudinal Factors (AF) on the relationship between Perceived Ease of Use (PEOU) and Behavioral Intention (BE) as shown in Figure 2. The interaction term (PEOU $\times$ AF) was significant and negative ($\beta = -0.206$, $p < .001$), confirming AF's moderating role. The conditional influence of PEOU on BE could be expressed as: Influence = $1.100 + (-0.206) \times \text{AF}$. This formula suggests that for every one-unit increase in AF, the positive impact of PEOU on BE decreases by 0.206. To further discuss this trend using simple slope analysis, the analysis has been compared with a low (M-1SD), moderate (M), and high (M + 1SD) level of AF. The results indicated that the relationship

between PEOU and BE was the greatest at low levels of AF (simple slope = 1.100, $p < .001$) and became steadily smaller with higher levels of AF (Table 9).

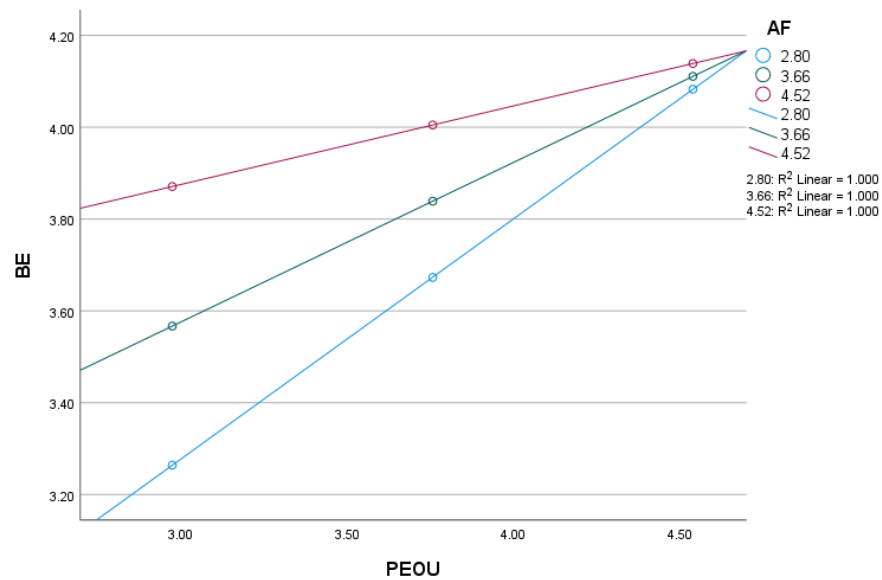


Figure 2

TABLE 9

Model	coeff	se	t	p	LLCI	ULCI
constant	-1.002	0.634	-1.579	0.115	-2.251	0.246
PEOU	1.100	0.179	6.135	0.000	0.747	1.453
AF	0.966	0.198	4.876	0.000	0.576	1.356
Int_1	-0.206	0.053	-3.868	0.000	-0.310	-0.101
PEOU	AF	BE				
2.855	2.668	3.150				
3.758	2.668	3.648				
4.660	2.668	4.146				
2.855	3.658	3.525				
3.758	3.658	3.839				
4.660	3.658	4.153				
2.855	4.648	3.901				
3.758	4.648	4.031				
4.660	4.648	4.161				

A similar analysis was conducted with CE as the outcome, as shown in Figure 3. The product term of PEOU and AF ($PEOU \times AF$) was still significant and negative ($\beta = -0.096$, $p = 0.008$). The conditional effect is as follows: $Effect = 0.526 + (-0.096) \times AF$, indicating that the slope of PEOU on CE decreases by 0.096 units with a one-unit increase in AF. This was confirmed by the results of the simple slope analysis, which indicated that the influence of PEOU on CE was most pronounced when AF was low and was lower when AF was higher (Table 10).

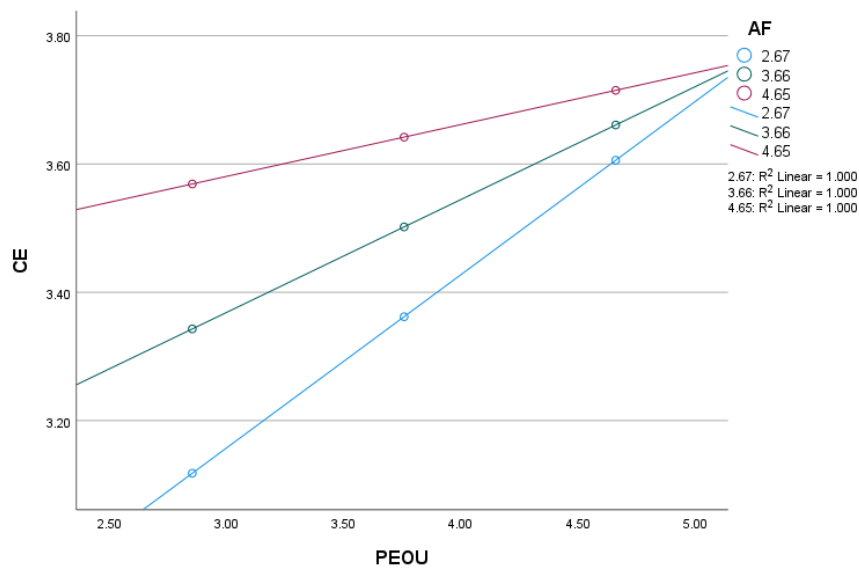


Figure 3

TABLE 10

Model	coeff	se	t	p	LLCI	ULCI
constant	1.009	0.428	2.359	0.019	0.167	1.85
PEOU	0.526	0.121	4.351	0.000	0.288	0.764
AF	0.501	0.134	3.751	0.000	0.238	0.764
Int_2	-0.096	0.036	-2.67	0.008	-0.166	-0.025
PEOU	AF	CE				
2.855	2.668	3.118				
3.758	2.668	3.362				
4.66	2.668	3.606				
2.855	3.658	3.343				
3.758	3.658	3.502				
4.66	3.658	3.661				
2.855	4.648	3.569				
3.758	4.648	3.642				
4.66	4.648	3.715				

In summary, AF consistently acts as an effective substitute moderator. The negative interaction terms coefficients in both BE and CE models suggest that the favorable effects of Perceived Ease of Use are strongest amongst users who initially had less favorable/neutral attitudes. In effect, it means that investing in and emphasizing a high level of ease of use of a system is an important approach toward generating positive behavioral intentions and experiences from users who are perhaps skeptical or not highly involved. Second, for highly positive users, the ease of use of the system is relevant but less determinative in influencing their behaviors.

The analysis examined the moderating effect of Behavioral Impact (BI) on Perceived Performance (PP) in relation to Behavioral Intention (BE), as shown in Figure 4. A strong negative interaction was found ($PP \times BI: \beta = -0.273, p < 0.001$). This association can be expressed as a formula: $\text{Effect} = 1.165 + (-0.273) \times BI$. This indicates a strong attenuation effect, where the slope of PP on BE decreases by 0.273 for every one-unit increase in BI. Simple slope analyses, graphed at low versus high levels of BI for PP (Fig. 1), supported the finding that the beneficial effect of PP on BE was significantly pronounced when BI was low and tended to attenuate as BI increased (Table 11).

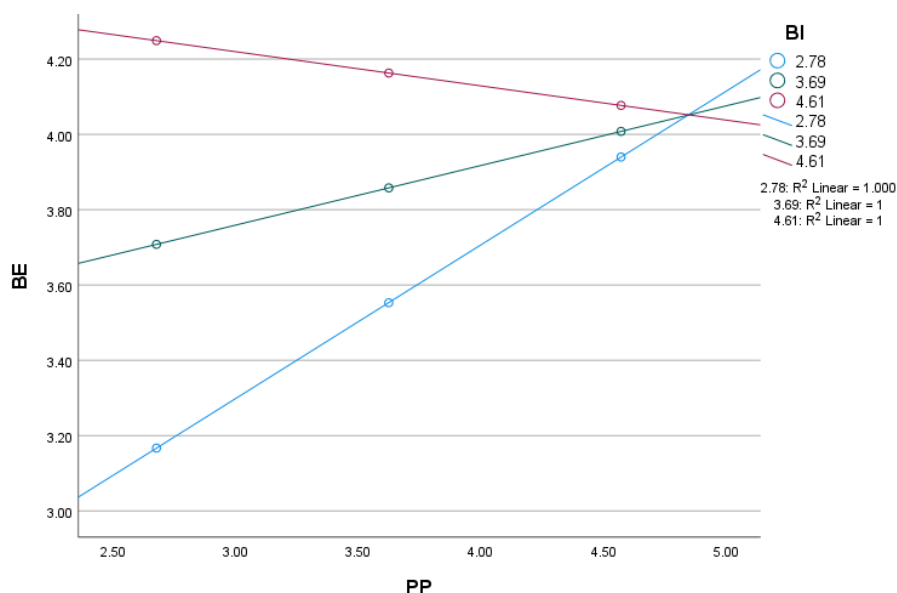


Figure 4

Model	coeff	se	t	p	LLCI	ULCI
constant	-1.591	0.599	-2.656	0.008	-2.769	-0.412
PP	1.165	0.179	6.506	0.000	0.812	1.517
BI	1.320	0.177	7.469	0.000	0.972	1.668
Int_3	-0.273	0.050	-5.445	0.000	-0.371	-0.174

PP	BI	BE
2.679	2.775	3.167
3.625	2.775	3.553
4.571	2.775	3.940
2.679	3.691	3.708
3.625	3.691	3.858
4.571	3.691	4.008
2.679	4.608	4.249
3.625	4.608	4.163
4.571	4.608	4.077

This moderation effect was also evident in the relationship between PP and CE, as shown in Figure 5. The interaction term ($PP \times BI$) was again negative and statistically significant ($p < 0.001$). The conditional effect is described by: $\text{Effect} = 0.632 + (-0.145) \times BI$, indicating that the strength of the relationship between PP and CE is also significantly weakened as BI increases. The pattern of simple slopes echoed the initial results and demonstrated a greater influence of PP on CE in low BI (Table 12).

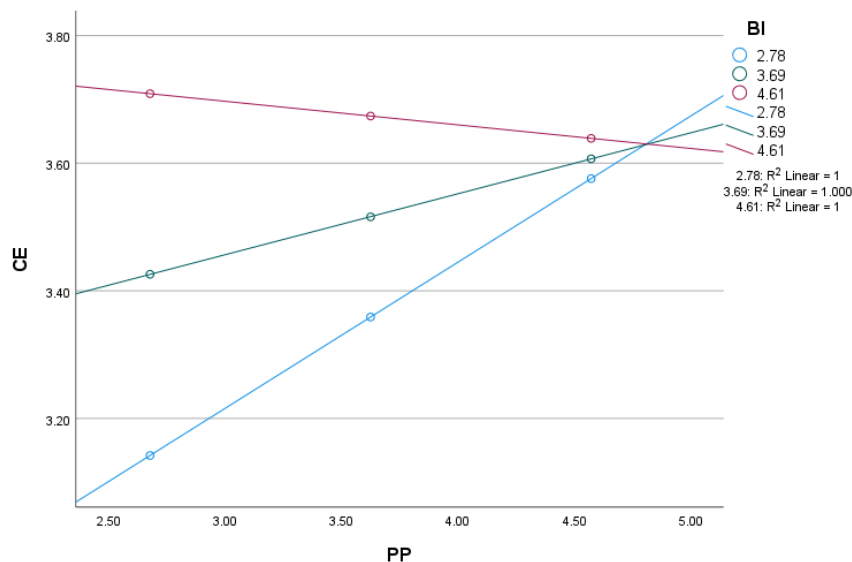


Figure 5

TABLE 12	Model	coeff	se	t	p	LLCI	ULCI
	constant	0.589	0.407	1.449	0.148	-0.211	1.389
	PP	0.632	0.122	5.204	0.000	0.393	0.872
	BI	0.699	0.120	5.826	0.000	0.463	0.935
	Int_4	-0.145	0.034	-4.279	0.000	-0.212	-0.079
	PP	BI	CE				
	2.679	2.775	3.142				
	3.625	2.775	3.359				
	4.571	2.775	3.576				
	2.679	3.691	3.426				
	3.625	3.691	3.516				
	4.571	3.691	3.607				
	2.679	4.608	3.709				
	3.625	4.608	3.674				
	4.571	4.608	3.639				

Parallel to AF, BI is a strong substitute moderator. The negative interaction effects with both the BE and CE models show that the impact of Perceived Performance is stronger on those subjects who have not yet been heavily influenced by other resources. Ultimately, high-performance perceptions are crucial for motivating the intentions and experiences of low-engaged or less engaged users. In contrast, among those who already have a high behavioral impact, the relative weight of performance perceptions on attitudinal and behavioral consequences is substantially less.

5. Discussion

Applying the Stimulus-Organism-Response (S-O-R) framework developed by (Jacoby, 2002) and (Eroglu et al., 2003), this research aims to examine empirically the effects of key characteristics of Generative AI (Stimuli): Perceived Ease of Use (PEOU), Perceived Interactivity(PI), and Perceived Personalization(PP), on Purchase Intention(Response) via intermediate variables concerning both Behavioral Engagement and Cognitive Engagement(Organism). The results are also highly supportive of the proposed hypotheses and thus have implications for extending the S-O-R framework to human-GenAI interaction in commerce, an emerging research domain (Gursoy et al., 2023; Kumar & Shankar, 2025).

Results indicate that all three GenAI attributes are significantly beneficial to the enhancement of both Behavioral Engagement (BE) and Cognitive Engagement (CE), providing evidence in support of H1a, H1b, H2b, H3a, and H3b. The research outcomes align with the existing literature, further supporting our model. Indeed, the positive effects of PEOU on engagement (H1a, H1b) resonate with central explanations proposed in the Technology Acceptance Model (TAM) (Davis, 1989) and is supported by more recent AI-specific literature that suggests a seamless user experience leads to lower cognitive load, freeing up mental resources for deeper immersion and exploration (Qian, 2025; S. Singh & Paiva, 2025). Likewise, strong influences of PI on BE and CE (H2a, H2b) are aligned with evidence in extant literature that suggests bidirectional, responsive, and synchronized interactions as the way to elicit user attention, trigger curiosity and intellectual interest, and motivate active participatory behavior (Agarwal et al., 2025; Joo & Yang, 2023). In addition, the substantial impact of PP on both engagement qualities (H3a, H3b) emphasized how powerful personalized experiences can be as a motivational factor. This was consistent with (Gao et al., 2023; Hu & Shao, 2025), when people feel that the information is personally tailored to them, it fulfills fundamental needs for content relevance and increases their commitment of cognitive and behavioral resources.

Importantly, the findings confirm BE and CE as central mediating factors (H4, H5, H6a-c). The substantial influence of BE (H4) and CE (H5) on Purchase Intention validates that engagement

is not a statefulness construct, but an activism of commercial results, as proposed in Consumer Engagement Theory (CET) (Agarwal et al., 2025; Hollebeek et al., 2023). The mediation analyses (H6a-c) consequently suggest that these internal psychological states primarily mediate the effects of PEOU, PI, and PP on Purchase intention. For example, partial mediation occurred between PEOU and PI, or between PI and PU, while BE completely mediated PP's effect on Purchase Intention. This is an important finding that extends beyond the results from (Teepapal, 2025), which showed personalization's effects on behavior are mediated by user engagement, by providing evidence for such a full mediation in the case of conversational GenAI. This implies that the personalized effect in GenAI is not automatically present but needs to be triggered by the user's succeeding behavioral events (e.g., exploring recommended products) (Gao et al., 2023).

The moderating effects provide important boundary conditions for refining the theoretical model. Unlike the initial expectation (H7b), AI Familiarity (AF) negatively moderated the relationships between PEOU and both BE and CE. This means that the positive impact of ease of use on performance is greater among users with low AI familiarity. For such users, getting them to use an easy-to-understand platform is critical because overcoming most initial resistance and "sense of falsity" towards Artificial Intelligence (AI) systems will lead them to engage (Aljarah et al., 2025). For highly experienced users accustomed to AI's capabilities and limitations (Arce-Urriza et al., 2025), they may come to take ease of use for granted, reducing its perceived relative importance. This discovery contributes to and extends the literature on technology adoption by revealing user familiarity as an influence that can mitigate the significance of a core TAM construct in developed adoption settings.

Conversely, as hypothesized (H7a), Brand Identification (BI) positively moderated the relationship between PP and engagement. The above perception aligns with pioneering research on brand relationships (Bhattacharya & Sen, 2004; Davvetas & Diamantopoulos, 2017), which suggests that BI goes beyond mere description of relationships and serves as a psychological extender. Consumers who already have a strong relationship with the brand think more personalized communication from their own brand's GenAI agent is more meaningful,

credible, and relevant (Ghorbanzadeh et al., 2025), which intensifies how they engage with it in comparison to those low in BI.

5.1 Theoretical Implications

The study enriches the current literature on AI-influenced consumer behavior by integrating the Technology Acceptance Model (TAM) with Consumer Engagement Theory (CE). It confirms the application of the S-O-R model in conversational GenAI (Erensoy et al., 2024; Premathilake et al., 2025), which drives behavior and contributes to a deeper understanding of how cognitive engagement and behavioral engagement work as two mediating variables. Furthermore, the moderating effects of brand identification and AI familiarity also respond to calls for more context- or person-specific research in AI (Foroughi et al., 2025; Kumar et al., 2025) as they offer a more nuanced understanding of the technology-to-consumer outcome process.

5.2 Managerial Implications

In the case of brands adopting GenAI, the revelations only enhance the significance of developing systems that are not only easy to use and responsive but also highly personalized. The effects of personalization will be reinforced by the creation of strong brand identities within the firms by the marketers (Ghorbanzadeh et al., 2025). The aspect of convenience can be a highly motivating incentive for even the non-expert population exposed to AI to a somewhat lesser degree. AI interaction training and transparency may reduce the “sense of falsity” (Aljarah et al., 2025) and improve perceived authenticity (Feng & Sun, 2025), increasing trust and conversion.

6. Conclusion

Generative AI (GenAI) plays an important role in digital marketing and customer service. It can increase the creation of highly customized and interactive customer experiences. The present research aims to investigate the relationships between the salient features of GenAI, perceived ease of Use, Perceived Interactivity, Perceived Personalization, and consumers' purchase intention. The research is anchored in the Stimulus-Organism-Response (S-O-R) model, where these features are postulated to function as stimuli to intensify Behavioral Engagement (BE) and Cognitive Engagement (CE), and to activate the respective mediators of both to purchase intentions. Additionally, the model aims to investigate the moderating roles of Brand Identification and AI Familiarity. An empirical study involving 320 consumers who participated in the GenAI chatbot in select economic regions of China was conducted to confirm the model. The data were subjected to SEM and moderation analysis. The findings reveal that the model's constituents, PEOU, PI, and PP, further promote BE and CE, which in turn fully or partially precipitate the mediation of the model in influencing purchase intention. Some of the findings are of interest, for example, that AI familiarity negatively moderates the relationship between perceived ease of use and engagement, which is contrary to the findings that Brand Identification positively moderates the relationship between personalized perception and engagement. The present research offers theoretical implications for the psychological dynamics underlying the adoption of GenAI within the framework of AI-based marketing theory.

6.1 Limitations

Despite its contributions, this study has limitations. First, the data were obtained in 4 economic regions of China, thus limiting the generalizability of the findings to other cultural or economic settings. Second, the cross-sectional nature of the data means that we cannot make causal claims or examine how these relationships evolve over time. Third, the research considered only behavioral and cognitive engagement, but not the affective engagement, which is regarded as a fundamental part of user experience.

6.2 Future Research Directions

This study has limitations that suggest promising prospects for future work. To enhance the external validity of the findings, it can be of use in later studies to replicate this model in other nations so that the cross-cultural generalizability can be measured (Gursoy et al., 2023). Longitudinal designs would enable researchers to observe the nature of these relationships' development as users gain increasing familiarity with GenAI, potentially demonstrating that the mediation and moderation effects evolve. Additionally, it would be necessary to add the emotional aspect of engagement (Hollebeek et al., 2023), which would help to understand the experience of GenAI among users better. Finally, exploring additional potential moderators, such as consumer privacy issues (Kronemann et al., 2023) or the type of product sold, may contribute to enlarging the image that we already have regarding the circumstances under which GenAI may be useful to the extent of altering the purchase intentions.

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QUESTIONNAIRE

CONSTRUCT	items	likter	reference
PERCEIVED EASE OF USE (PEOU)	PEOU1: Using generative AI is innovative compared to other tools I use for shopping	1.Strongly Disagree	(Davis, 1989; Singh & Paiva, 2025)
	PEOU2: Using generative AI offers new and creative ways to support my shopping decisions	2. Disagree 3. Neutral	
	PEOU3: Using generative AI in my shopping process is a groundbreaking idea	4. Agree 5. Strongly	
	PEOU4: Using generative AI enhances my shopping experience	Agree	
PERCEIVED INTERACTIVITY (PI)	PI1: Generative AI chatbot were very slow in responding to my request	1.Strongly Disagree	(Gao et al., 2023)
	PI2: The generative AI chatbot can accurately provide me with the product	2. Disagree 3. Neutral	
	PI3: When I encounter a shopping-related problem, the generative AI chatbot can provide me with a solution	4. Agree 5. Strongly	
	PI4: The generative AI chatbot can effectively collect customer feedback	Agree	
	PI5: The generative AI chatbot makes me feel like I'm having a real conversation		
	PI6: The generative AI chatbot does not encourage users to talk back		
PERCEIVED PERSONALIZATION (PP)	PP1: The generative AI chatbot stores my preferences and offers me extra services based on my preferences	1.Strongly Disagree	(Gao et al., 2023)
	PP2: The generative AI chatbot does a pretty good job guessing what kinds of products I might want and making suggestions	2. Disagree 3. Neutral 4. Agree	

	PP3: The generative AI chatbot know what I want	5. Strongly	
	PP4: The generative AI chatbot setup can be personalized to my needs	Agree	
	PP5: The service provided by the generative AI chatbot is customized exactly to my question		
BEHAVIORAL ENGAGEMENT (BE)	BE1: I often actively engage with the generative AI chatbot during my shopping	1.Strongly Disagree	(Agarwal et al., 2025;
	BE2: I frequently explore new products, activities, or promotions by using the generative AI chatbot	2. Disagree 3. Neutral	Fang et al., 2017;
	BE3: I explore and interact with various features and recommendations provided by the generative AI chatbot.	4. Agree 5. Strongly Agree	McLean & Wilson, 2019)
COGNITIVE ENGAGEMENT (CE)	CE1: I am highly focused on my interactions with the generative AI chatbot	1.Strongly Disagree	(Agarwal et al., 2025;
	CE2: I often find myself engaged in problem-solving or decision-making processes with the help of the generative AI chatbot	2. Disagree 3. Neutral 4. Agree	McLean & Wilson, 2019;
	CE3: I generally do not need to invest a lot of mental effort in processing and understanding the information provided by the generative AI chatbot	5. Strongly Agree	O'Brien et al., 2018)
	CE4: I feel that my interactions within the generative AI chatbot contribute to my shopping decisions or purchasing knowledge		
	CE5: I find myself deeply engaged for a long duration when using the generative AI chatbot in a single session		
AI FAMILIARITY (AF)	AF1: I have had experience interacting with AI	1.Strongly	(Feng & Sun,
	AF2: I am familiar with AI-generated content (texts, audio, visuals, etc.)	Disagree 2. Disagree	2025)

	AF3: I am familiar with AI-powered applications or tools in my daily life (e.g. voice assistants, recommendation systems). AF4: I feel comfortable discussing the capabilities and limitations of AI.	3. Neutral 4. Agree 5. Strongly Agree	
BRAND IDENTIFICATION (BI)	BI1: Through interactions with the Gen AI chatbot, I feel a strong sense of belonging to this brand. BI2: The Gen AI chatbot helps me identify strongly with this brand. BI3: Through the Gen AI chatbot, I can believe that this brand embodies. BI4: Due to the Gen AI chatbot, this brand is like a part of me. BI5: Through the Gen AI chatbot, this brand has a great deal of personal meaning to me.	1.Strongly Disagree 2. Disagree 3. Neutral 4. Agree 5. Strongly Agree	(Davvetas & Diamantopoulou, 2017; Stokburger-Sauer et al., 2012)
PURCHASE INTENTION (PINT)	PINT1: I would purchase products or services through Gen AI chat PINT2: The probability that I would consider buying through Gen AI chat is high PINT3: My willingness to purchase through Gen AI chat is strong	1.Strongly Disagree 2. Disagree 3. Neutral 4. Agree 5. Strongly Agree	(Arachchi & Samarasinghe, 2024; Prentice et al., 2019)